Kentaro Mori

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EDUCATION

Doctor of Philosophy in Civil Engineering Expected August 2027
The University of Texas at Austin (Austin, TX) 4.0 GPA
Advisor: Kara Kockelman https://www.caee.utexas.edu/prof/kockelman/

Master of Engineering in Civil Engineering, Global Environmental Leaders Program March 2022
Nagoya University (Nagoya, Japan) 4.0 GPA

Bachelor of Science in Civil Engineering, minor in mathematics, *summa cum laude* May 2019
Lafayette College (Easton, PA) 3.88 GPA

RESEARCH EXPERIENCE

Graduate Research Assistant June 2023 – Present
UT Austin Department of Civil, Architectural, and Environmental Engineering
- Investigated the effects of spatial resolution on shared autonomous vehicle simulations.
- Analyzed travel demand variations throughout the year in Texas using emerging data sources.
- Evaluated congestion pricing schemes using the POLARIS transportation simulator.
- Designed a novel autonomous transit system and implemented it in POLARIS.

Research Assistant September 2022 – January 2023
MIT Intelligent Transportation Systems Lab
- Developed a freight model for the Greater Toronto and Hamilton Area to be used in the SimMobility transportation simulator.

Research Associate April 2022 – August 2022
Nagoya University Transport and Environmental Dynamics
- Contributed to projects estimating latent demand in various contexts, such as for street-hailing taxis from GPS trajectory data and of senior citizens in mountainous areas.

Data Science Summer Institute Intern June 2021 – September 2021
Lawrence Livermore National Laboratory (remote)
- Developed a simulation to forecast the spatiotemporal distribution of the charging demand of shared autonomous vehicles.
- Analyzed the effect of time-of-use electricity pricing schemes on the charging demand of shared autonomous vehicles in Austin, TX.

Graduate Researcher October 2019 – March 2022
Nagoya University Transport and Environmental Dynamics Lab
- Analyzed effects of autonomous taxis on mode choice and traffic conditions in Nagoya.
- Performed mode choice analysis, using both revealed and stated preference data.
- Developed a combined mode split-assignment model based on user equilibrium in Fortran.

Undergraduate Research Assistant August 2018 – May 2019
Lafayette College Concrete Laboratory
- Designed precast concrete bridge deck panel specimens to verify PennDOT standard joint detail and consider improvements using ultra-high performance concrete.
- Designed a steel load application frame to be used for deck panel strength testing.
- Designed and constructed formwork for concrete pedestal.

Undergraduate Research Assistant August 2016 – May 2018
Lafayette College Department of Civil and Environmental Engineering
- Explored the feasibility of using biofilm-forming bacteria to reduce permeability in sands.
- Developed a method for reliably measuring permeability change in sand columns.
- Performed permeability tests to determine the effect of biofilm-forming bacteria on sands.

PUBLICATIONS

1. **Mori, K.**, de Souza, F., Gurumurthy, K.M., Auld, J., and Kockelman, K.M. (2026): Comparing Delay-, Distance-, and Cordon-Based Congestion Pricing Strategies via Large-Scale Simulation. ***Transportation Research Part A***, Vol. 208, 104959. (https://doi.org/10.1016/j.tra.2026.104959)
2. **Mori, K.**, Fakhrmoosavi, F., Gurumurthy, K.M., Camargo, P., and Kockelman, K.M. (2026): Impacts of Spatial Resolution on Agent-based Transportation Simulations with Shared Autonomous Vehicles. ***Computers, Environment and Urban Systems***, Vol. 124, 102371. (https://doi.org/10.1016/j.compenvurbsys.2025.102371)
3. **Mori, K.**, Ayebare, P.A., Gurumurthy, K.M., and Kockelman, K.M. (2026): Feast or Famine: Fleet Profitability and Other Performance Metrics Across Days of the Year, *Under review for publication in* ***Transportation Policy and Technology***.
4. **Mori, K.**, Ayebare, P.A., Gurumurthy, K.M., Kockelman, K.M., and Huang, Y. (2026): Shared Autonomous Vehicle Operations Across Distinct Dallas-Fort Worth Geofences. *Working paper*.
5. **Mori, K.**, Gurumurthy, K.M., and Kockelman, K.M. (2026): Subsidizing Shared Autonomous Vehicle Operators to Serve Bus Riders: Results from Regional Simulations. *Working paper.*
6. **Mori, K.** and Kockelman, K.M. (2024): Day-of-Week, Month, and Seasonal Demand Variations: Comparing Flow Estimates Across New Travel Data Sources, ***Findings***. (https://doi.org/10.32866/001c.118815)
7. **Mori, K.**, Miwa, T., Abe, R. and Morikawa, T. (2022): Equilibrium Analysis of Trip Demand for Autonomous Taxi Services in Nagoya, Japan, ***Transportation Research Part A***, Vol. 166, pp. 476–498. (https://doi.org/10.1016/j.tra.2022.10.009)

PRESENTATIONS & PROCEEDINGS

1. **Mori, K.**, Gurumurthy, K.M., Jayashankar, N., and Kockelman, K.M.: Subsidizing Shared Autonomous Vehicle Operators to Serve Bus Riders: Results from Regional Simulations. *106th Transportation Research Board (TRB) Annual Meeting*, Washington D.C., January 2027
2. **Mori, K.**, Gurumurthy, K.M., and Kockelman, K.M.: Robotaxis as Transit: Simulation-Informed Benefit-Cost Comparisons. *105th Transportation Research Board (TRB) Annual Meeting*, Washington D.C., January 2026. *8th Bridging Transportation Researchers Online Conference*, August 2026.
3. **Mori, K.**, de Souza, F., Gurumurthy, K.M., Auld, J., and Kockelman, K.M.: Comparing Delay-, Distance-, and Cordon-Based Congestion Pricing Strategies via Large-Scale Simulation. *104th Transportation Research Board (TRB) Annual Meeting*, Washington D.C., January 2025. *71st North American Meetings of the Regional Science Association International*, New Orleans, LA, November 2024. *6th Bridging Transportation Researchers Online Conference*, August 2024.
4. **Mori, K.** and Kockelman, K.M.: Day-of-Week, Month, and Seasonal Demand Variations: Comparing Flow Estimates Across New Travel Data Sources, *104th Transportation Research Board (TRB) Annual Meeting*, Washington D.C., January 2025. *71st North American Meetings of the Regional Science Association International*, New Orleans, LA, November 2024. *6th Bridging Transportation Researchers Online Conference*, August 2024.
5. **Mori, K.**, Fakhrmoosavi, F., Gurumurthy, K.M., Camargo, P., and Kockelman, K.M.: Impacts of Spatial Resolution on Agent-based Transportation Simulations with Shared Autonomous Vehicles, *6th Bridging Transportation Researchers Online Conference*, August 2024. *17th International Conference on Travel Behavior Research*, Vienna, Austria, July 2024. *103rd Transportation Research Board (TRB) Annual Meeting*, Washington D.C., January 2024.
6. Goto, D., Miwa, T. **Mori, K.**, and Morikawa, T.: A Preliminary Study on Traffic Assignment Model with Complex Travel Mode Choices, *Proceedings of Infrastructure Planning* 63, June 2021. (Japanese)
7. **Mori, K.**, Miyoshi, I, Miwa, T., and Morikawa, T.: Equilibrium Analysis of Trip Demand of Autonomous Taxi Services in Nagoya, Japan, *Proceedings of Infrastructure Planning* 62, Shinshu University (Online), November 2020. (Japanese)

PEER REVIEW SERVICE

ACM Transactions on Cyber-Physical Systems, Bridging Transportation Researchers Conference, Computers and Operations Research, Journal of Advanced Transportation, Journal of Transport Geography, Research in Transportation Economics, Transactions in GIS, Transportation, Transportation Planning and Technology, Transportation Research Board (TRB) Annual Meeting, Transportation Research Part A: Policy and Practice, Transportation Research Part E: Logistics and Transportation Review, Transportation Research Record, Travel Behaviour and Society

TEACHING AND MENTORING EXPERIENCE

Teaching Assistant, CE311S Probability and Statistics for Civil Engineers (UT Austin) Spring 2026
- Led laboratory sessions to reinforce students' understanding of statistical concepts.

Teaching Assistant, CE392H Regional Transportation Planning (UT Austin) Spring 2025
- Instructed students on TransCAD, graded homework, and advised final projects.

Undergraduate Thesis Mentor Fall 2024 – Spring 2025
- Instructed undergraduate student with use of GIS software and transportation data.

Teaching Assistant, CE367G Ground-Based Transportation Systems (UT Austin) Spring 2024
- Led laboratory sessions, hosted office hours, created homework assignments, and graded.

International Student Tutor (Nagoya University) Fall 2020 – Fall 2022
- Provided tutoring to international students in engineering courses and Japanese language.

Teaching Assistant, Introduction to Engineering (Lafayette College) Fall 2018
- Set up and ran structural engineering demonstrations and experiments.

Teaching Assistant, Strength of Materials (Lafayette College) Spring 2018
- Provided group tutoring sessions twice a week.

LEADERSHIP AND SERVICE

Student Transportation Council Executive Director and ITE President 2025-Present
UT Austin
- Lead chapter operations and organized seminars and networking events for transportation students.

Bridging Transportation Researchers Conference
Organizing Committee Member 2025-Present
Technical Lead 2023-Present
- Oversaw virtual session logistics, trained volunteers, and ensured smooth technical operations.

Curriculum Writer for US Academic Decathlon 2025
- Developed high school-level materials on transport economics for national competition study guides.

HONORS AND AWARDS

Cockrell Fellowship, UT Austin 2023-Present
- $6,000/year for 4 years

Professional Development Award, UT Austin 2024

Japan Student Services Organization Student Exchange Support Program 2022-2023
- Scholarship from the Japanese government to pursue a graduate degree abroad

Marquis Scholarship, Lafayette College 2015-2019
- Half-tuition merit scholarship to attend Lafayette College

EXCEL Scholarship 2016-2018
- Scholarship to engage in collaborative research projects with Lafayette faculty

CaPA Scholarship 2016-2019
- Scholarship to contribute to the Lafayette College community through the arts

Delaware Valley Geo-Institute Scholarship 2018

Phi Beta Kappa – Liberal arts honor society 2018-Present

Tau Beta Pi – Engineering honor society 2018-Present

Pi Mu Epsilon – Mathematics honor society 2018-Present

Order of Omega – Greek life honor society 2018-Present

TECHNICAL SKILLS

- Programming languages: Python, C++, Java, Julia, Fortran, R, Matlab, GAUSS, VBA, SQL
- Other: Linux, ArcGIS, QGIS, TransCAD

LANGUAGES

English (Native), Japanese (Native)

REFERENCES

Kara Kockelman, PhD, PE Professor, UT Austin kkockelm@mail.utexas.edu 512-471-0210	Krishna Murthy Gurumurthy, PhD Argonne National Laboratory kgurumurthy@anl.gov 630-252-1643	Stephen Boyles, PhD, PE Professor, UT Austin sboyles@austin.utexas.edu 512-471-3548